A Mechanistic Model for Radiation-Induced Segregation Controlled Grain-Boundary Oxidation in Austenitic Stainless Steels

Zhuo Wang Md Shahrier Hasan Jaime Marian

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Abstract

Irradiation accelerates the intergranular oxidation that precedes irradiation-assisted stress corrosion cracking of austenitic stainless steels in PWR primary water, but no model connects the radiation-induced segregation (RIS) at a grain boundary to the kinetics of its oxidation. We develop a model that couples RIS to grain-boundary oxidation through the oxide itself: selective interfacial reactions convert the local, segregated alloy composition into an oxide composition, the oxide retains that composition as it grows, and the oxygen diffusivities along and across the boundary are computed from the resulting composition profile, with the porosity left by the dissolution of oxidized Si included. With one parameter set for all doses, the model reproduces the intergranular penetration depths measured on proton-irradiated 304 nuclear-grade stainless steel after 500 h at 0, 0.5 and 3 dpa (45, 69 and 106 nm calculated against 40, 62 and 105 nm measured), and it predicts the Cr minimum observed at the centre of the oxide of the irradiated specimens. Separating the two segregation routes shows that Si enrichment accounts for most of the dose dependence and Cr depletion for a minor part. Applied without adjustment to concurrent irradiation and oxidation of a type 316 baffle-former bolt after 40 years of service, the model gives a penetration depth of 2.6 µm, within the 2 to 3 µm measured.

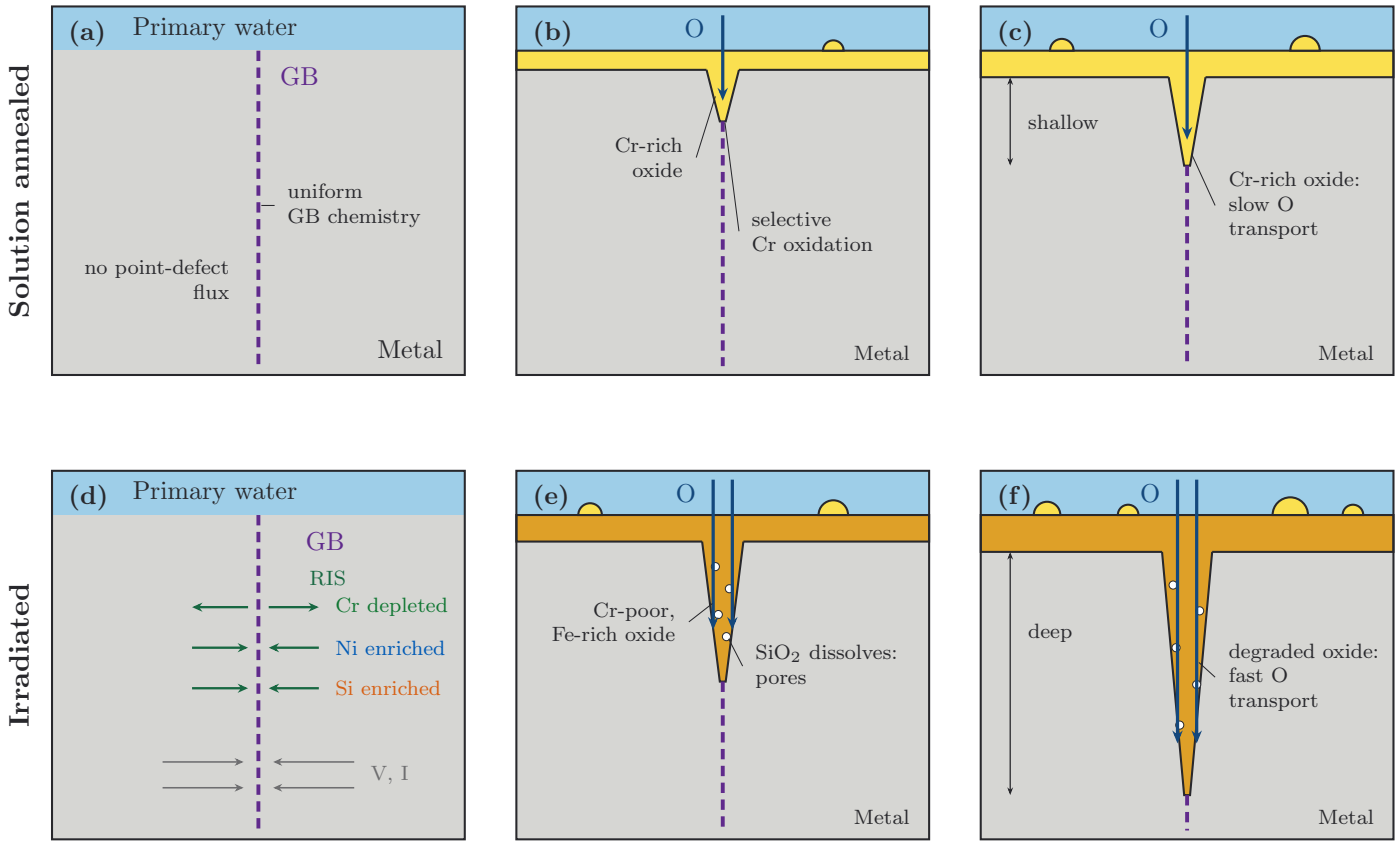


Figure 1: Physical picture of intergranular oxidation in primary water without (a–c) and with (d–f) prior irradiation. (a) Solution-annealed boundary with uniform chemistry. (b) Oxygen entering at the boundary mouth oxidizes Cr selectively and forms a Cr-rich intergranular oxide. (c) The Cr-rich oxide limits oxygen transport along the boundary and the penetration stays shallow. (d) Irradiation drives point defects to the boundary and the resulting radiation-induced segregation depletes Cr and enriches Ni and Si. (e) On exposure the Cr-depleted boundary forms a Cr-poor, Fe-rich oxide, and the segregated Si oxidizes and dissolves, leaving pores. (f) The degraded oxide transports oxygen faster along the boundary and the oxide penetrates deeper.

1 Introduction

Irradiation-assisted stress corrosion cracking (IASCC) is a major degradation mode of austenitic stainless steel core internals in light-water reactors [39]. In irradiated stainless steels exposed to high-temperature water, intergranularoxidation weakens grain boundaries and precedes intergranular crack initiation [38], and grain-boundary oxidation has been identified as a necessary precursor of IASCC at low doses [27]. In 304 stainless steel (304SS) exposed to pressurized water reactor (PWR) primary water, the extent of this oxidation depends on the chemical changes introduced by prior irradiation. Deng et al. [11] found that increasing the proton irradiation dose produced deeper intergranular oxidation in 304 nuclear-grade stainless steel after water exposure, accompanied by a decrease in the Cr content of the intergranular oxide. Cr depletion is not the only chemical change that matters. Using model alloys with Si and Cr contents chosen to mimic the segregated boundary, Du et al. [13] showed that Si enrichment promotes intergranular oxidation and cracking even when Cr is not depleted: Si forms SiO_2 that is unstable in high-temperature water and dissolves preferentially, leaving a porous oxide that provides pathways for oxygen diffusion along grain boundaries. These observations connect irradiation to both the chemistry and the penetration depth of the intergranular oxide. Turning them into a predictive relationship requires a description of how the changing oxide composition and structure affect oxygen transport and, consequently, the advance of the oxidation front.

Radiation-induced segregation (RIS) establishes the local alloy composition from which the intergranular oxide develops. Coupling between solute and irradiation-generated point-defect fluxes redistributes alloying elements near grain boundaries. Measurements on irradiated 304 nuclear-grade stainless steel reveal Cr depletion and Ni and Si enrichment at the grain boundary [9]. Subsequent oxidation selectively consumes alloying elements, changing both the oxide composition and the adjacent metal, where Ni enrichment has also been observed [9]. The lower Cr content of the oxide formed on irradiated boundaries has been associated with reduced resistance to oxygen diffusion [11].

Si oxidation and dissolution alter this transport resistance by changing the structure through which oxygen moves. The role of Si was established by Du et al. [13] with model alloys of systematically varied Cr and Si contents in hydrogenated water, which allowed the contributions of Cr depletion and Si

enrichment to be separated. They attributed the greater oxide degradation and cracking susceptibility at higher Si content to dissolution of SiO_2 in the inner oxide and the resulting porous, defective film, and their STEM analysis of intergranular oxides indicated faster grain-boundary diffusion at higher Si content. Related microscopy on proton-irradiated 316L identified Si enrichment at oxide tips and porosity behind them, consistent with preferential Si oxidation followed by dissolution [27]. Although that alloy differs from 304SS in bulk composition, the sequence itself is not alloy-specific: oxidation and dissolution of segregated Si appears as a step in the IASCC mechanism summarized by Was et al. [38] for austenitic stainless steels generally. Together, these studies provide a basis for including dissolution-generated porosity in the transport description. With the dependence on Cr content, they motivate representing the evolving oxide composition and porosity through effective oxygen diffusivities computed from both, which connect the segregation of multiple elements to the quantities governing oxygen transport.

Each half of this sequence has been modeled separately, and the two halves are not equally developed. The segregation half is settled enough that the grain boundary composition can be computed rather than only measured. Since the inverse-Kirkendall and rate-theory descriptions of RIS in concentrated alloys [30, 40], the composition profile at a grain boundary has been obtained from the coupled fluxes of point defects and alloying elements, with Fe–Cr–Ni parameterizations that reproduce the measured dose and temperature dependence of Cr depletion and Ni enrichment [1]. Current models go further and resolve the sink itself, predicting segregation at grain boundaries and dislocation cell walls [25] and the conditions under which Cr switches from enrichment to depletion [32]. All of these models return the segregation profile as their final output; none includes how the modified boundary composition affects the subsequent grain-boundary oxidation.

On the oxidation side, the dependence of intergranular penetration depth on dose and exposure time has been characterized experimentally [10, 11], but existing kinetic models of grain-boundary oxidation do not respond to the grain boundary chemistry. Models that resolve oxygen transport along the grain boundary treat the oxygen diffusivity as a single constant: in analytical or semi-analytical solutions that give the penetration depth in closed form [14, 16, 41], in finite-element descriptions in which it is fitted to measured oxide depths through a parabolic law [26, 44] or prescribed together with a constant reaction rate [3], and in phase-field simulations of a single stoichiometric chromia intrusion [12, 34]. In none of these does the diffu-

sivity depend on the composition of the oxide that has formed. Where the oxide identity does control transport, in multilayer descriptions of planar scales [21, 23] and in the mechanistic treatment of uniform corrosion in high-temperature water [35], the controlling phase is fixed at the outset rather than selected by the local alloy composition. The boundary composition enters only after conversion into an empirical parameter, as a Cr-content-dependent dissolution rate in sensitized steels [33] or as a sensitization index that increases with neutron fluence in the slip-oxidation prediction of crack growth [17]; crack-growth models that require a grain-boundary oxidation law borrow rate constants measured on bulk surfaces [37]. Thermodynamic selection of the oxide phases from the local alloy composition has been implemented for a Ni-base superalloy [15], but with a constant oxygen diffusivity, the bulk composition at the boundary, and without the Cr oxide that limits transport. No model takes the irradiation-modified boundary composition as input and lets the resulting oxide composition set the oxygen transport that governs further penetration.

To our knowledge, no model connects the segregated composition to the advance of the oxidation front through a transport quantity computed from both. Establishing this connection requires tracking how the segregated alloy supplies the interfacial reactions, how those reactions change the oxide composition, and how the resulting diffusivity affects subsequent oxygen delivery. Segregation profiles, oxide composition, interfacial Ni enrichment, and penetration depth then provide complementary observables for assessing the coupling.

Here, we develop a coupled RIS and grain-boundary oxidation model for 304SS, represented by the Fe–Cr–Ni–Si system. The coupling is made through the oxide itself: selective interfacial reactions convert the local, segregated alloy composition into an oxide composition, the oxide retains that composition as it grows, and the oxygen diffusivities along and across the boundary are computed from the resulting composition profile. Within this description, increasing the Cr content of the oxide lowers the diffusivity, whereas Si oxidation followed by complete dissolution contributes porosity that enhances transport. The same growth record gives the diffusivity for oxygen transport along the grain boundary and that for oxygen delivery across the oxide to the metal interface. The model therefore connects RIS to oxidation kinetics through the resulting change in oxygen transport, with metal consumption feeding back on the local composition. The complete-dissolution treatment is a simplification motivated by the mechanism dis-

cussed by Du and co-workers [13]. We examine the dose and exposure-time dependence of oxide penetration under sequential irradiation and water exposure, using separate calibration of the RIS and oxidation components and a common set of oxidation parameters across doses. The same equations, with irradiation and oxidation acting together, are then applied without adjustment to a baffle-former bolt after 40 years of service and compared with the depth of intergranular oxidation measured on it.

2 Model Formulation and Calibration

The model consists of two components: a radiation-induced segregation (RIS) model for the evolving GB chemistry and a reduced-dimensional oxidation model for oxygen transport, consumption, and oxide growth. The RIS-predicted Cr, Ni, Si, and Fe concentrations provide the local alloy composition for the oxidation model. Oxygen transport along the boundary is solved for the cross-section-averaged concentration of the oxidized zone, and the delivery of oxygen to the oxide/metal interface is closed by the fundamental mode of the steady diffusion equation in the cross-section. The oxide composition, recorded across the section as the oxide grows, determines the effective oxygen diffusivities along and across the boundary, which control the local oxygen consumption and oxide growth rate. The formulations of the two components are described below.

Figure 2 summarises the model. The computational domain, Fig. 2a, is a grain boundary that opens onto the surface exposed to the primary water. Oxygen enters at the mouth through the surface film and migrates along the boundary within the oxidized zone, whose half-width grows as the oxide thickens. Only the half-domain on one side of the boundary is simulated, the other half following by symmetry; it extends 50 nm normal to the boundary and 300 nm along it, on a mesh of 1 nm spacing.

The boundary acts as a sink of finite efficiency for the vacancies and self-interstitials produced by irradiation, Fig. 2b. The resulting redistribution depletes Cr and enriches Ni and Si at the boundary, and the imbalance between the two defect fluxes gives the segregation contribution $\mathbf{v}_{\mathrm{lat}}^{\mathrm{RIS}} = \mathbf{J}_{\mathrm{V}} - \mathbf{J}_{\mathrm{I}}$ to the lattice velocity; the profiles shown are calculated, mirrored about the boundary and convolved with a 1 nm FWHM Gaussian. At a given depth, Fig. 2c, the oxygen delivered along the boundary reaches the advancing oxide/metal interface through the oxide cross-section and is partitioned there

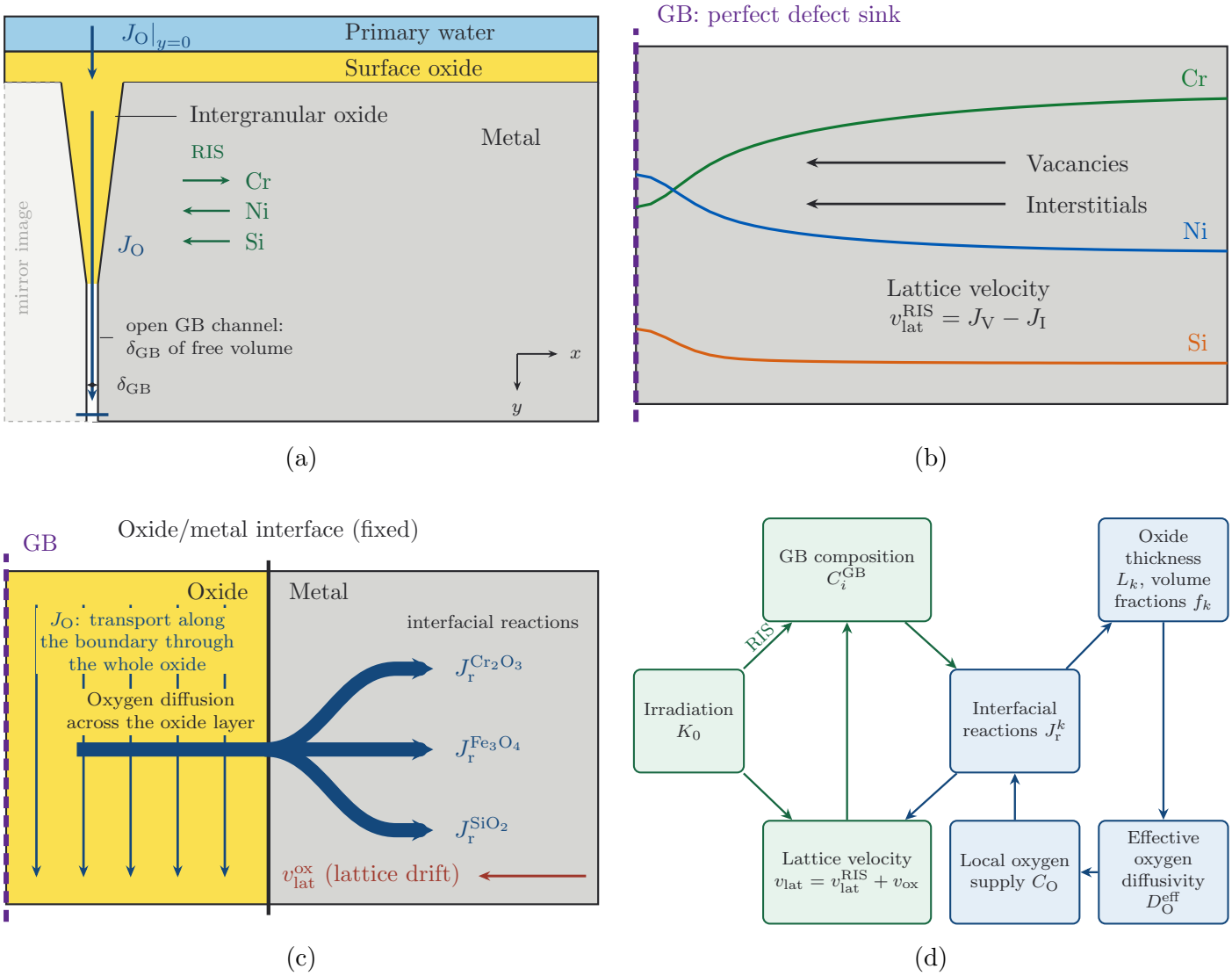


Figure 2: Overview of the model: (a) physical picture and computational domain, (b) radiation-induced segregation, (c) oxygen transport across the oxide and the interfacial reactions, and (d) coupling between the two components.

among the three oxides; the metal swept by the front is either oxidized, rejected into the metal, or carried away through the oxide, and the net removal sets the oxidation contribution v_{ox}. The two components are coupled as shown in Fig. 2d: irradiation fixes the boundary chemistry and part of the lattice velocity, the boundary chemistry controls the interfacial reaction rates, and the oxide that forms feeds back on the oxygen supply through its thickness and its effective diffusivity.

Notation. Alloying-element indices are $i, j \in \{\mathrm{Cr}, \mathrm{Ni}, \mathrm{Si}, \mathrm{Fe}\}$; oxide-phase indices are $k \in \{\mathrm{Cr_2O_3}, \mathrm{Fe_3O_4}, \mathrm{SiO_2}\}$. Superscripts V and I denote quantities associated with vacancies and self-interstitials; the superscript GB denotes a value taken on the grain boundary. C is a site fraction, $\mathbf{J}$ a flux in the two-dimensional metal domain, and J a flux on the one-dimensional grain-boundary line. The coordinate x is normal to the boundary and y runs along it; $\mathbf{n}$ is the unit normal to the boundary plane pointing from the metal towards the boundary, the outward normal of the metal domain, so that a flux $\mathbf{J} \cdot \mathbf{n}$ is counted positive towards the boundary.

2.1 Radiation-Induced Segregation Model

The RIS model is based on solving a set of coupled mass-conservation equations for the alloying elements (Cr, Ni, Si, and Fe) and point defects (vacancies and self-interstitials). For each alloying element i, the concentration evolves according to

$$\frac{\partial C_i}{\partial t} = -\nabla \cdot \mathbf{J}_i. \tag{1}$$

The site fractions of vacancies and self-interstitials are governed separately by

$$\frac{\partial C_{\mathrm{V}}}{\partial t} = -\nabla \cdot \mathbf{J}_{\mathrm{V}} + \epsilon K_0 - R\, C_{\mathrm{I}} C_{\mathrm{V}}, \tag{2}$$

$$\frac{\partial C_{\mathrm{I}}}{\partial t} = -\nabla \cdot \mathbf{J}_{\mathrm{I}} + \epsilon K_0 - R\, C_{\mathrm{I}} C_{\mathrm{V}}, \tag{3}$$

where C_{V} and C_{I} are the vacancy and self-interstitial site fractions, ϵ is the damage efficiency, K_0 is the radiation dose rate, and R is the vacancy–interstitial recombination rate coefficient. The flux of each alloying element

carried by each defect mechanism is

$$\mathbf{J}_i^{\mathrm{V}} = -\sum_j L_{ij}^{\mathrm{V}} \nabla(\mu_j - \mu_{\mathrm{V}}) \,, \tag{4}$$

$$\mathbf{J}_i^{\mathrm{I}} = -\sum_j L_{ij}^{\mathrm{I}} \nabla(\mu_j + \mu_{\mathrm{I}}) \,, \tag{5}$$

with the Onsager coefficients

$$L_{ii}^{\mathrm{V}} = \frac{C_i D_i^{\mathrm{V}} C_{\mathrm{V}}}{k_B T} \left(1 + \frac{1 - f_0^{\mathrm{V}}}{f_0^{\mathrm{V}}} \frac{C_i D_i^{\mathrm{V}}}{\sum_m C_m D_m^{\mathrm{V}}}\right), \tag{6}$$

$$L_{ij}^{\mathrm{V}} = L_{ji}^{\mathrm{V}} = \frac{1 - f_0^{\mathrm{V}}}{f_0^{\mathrm{V}}} \frac{C_i D_i^{\mathrm{V}} \, C_j D_j^{\mathrm{V}} \, C_{\mathrm{V}}}{k_B T \sum_m C_m D_m^{\mathrm{V}}}, \qquad i \neq j, \tag{7}$$

$$L_{ii}^{\mathrm{I}} = \frac{C_i D_i^{\mathrm{I}} C_{\mathrm{I}}}{k_B T} \left(1 + \frac{1 - f_0^{\mathrm{I}}}{f_0^{\mathrm{I}}} \frac{C_i D_i^{\mathrm{I}}}{\sum_m C_m D_m^{\mathrm{I}}}\right), \tag{8}$$

$$L_{ij}^{\mathrm{I}} = L_{ji}^{\mathrm{I}} = \frac{1 - f_0^{\mathrm{I}}}{f_0^{\mathrm{I}}} \frac{C_i D_i^{\mathrm{I}} \, C_j D_j^{\mathrm{I}} \, C_{\mathrm{I}}}{k_B T \sum_m C_m D_m^{\mathrm{I}}}, \qquad i \neq j, \tag{9}$$

where D_i^{V} and D_i^{I} are the mechanism-specific diffusion coefficients of element i, and f_0^{V} and f_0^{I} are the corresponding correlation factors. The net defect fluxes follow from conservation of lattice sites,

$$\mathbf{J}_{\mathrm{V}} = -\sum_i \mathbf{J}_i^{\mathrm{V}}, \tag{10}$$

$$\mathbf{J}_{\mathrm{I}} = \sum_i \mathbf{J}_i^{\mathrm{I}}, \tag{11}$$

and give rise to a lattice velocity

$$\mathbf{v}_{\mathrm{lat}}^{\mathrm{RIS}} = \mathbf{J}_{\mathrm{V}} - \mathbf{J}_{\mathrm{I}}. \tag{12}$$

Grain-boundary sink condition. The boundary is treated as a sink of finite efficiency for both point defects. At the boundary plane the defect fluxes obey the Robin conditions

$$\mathbf{J}_{\mathrm{V}} \cdot \mathbf{n}\big|_{x=0} = k_{\mathrm{V}}^{\mathrm{GB}} \left(C_{\mathrm{V}}\big|_{x=0} - C_{\mathrm{V}}^{\mathrm{eq}}\right), \tag{13}$$

$$\mathbf{J}_{\mathrm{I}} \cdot \mathbf{n}\big|_{x=0} = k_{\mathrm{I}}^{\mathrm{GB}} \left(C_{\mathrm{I}}\big|_{x=0} - C_{\mathrm{I}}^{\mathrm{eq}}\right), \tag{14}$$

where $C_{\mathrm{V}}^{\mathrm{eq}}$ and $C_{\mathrm{I}}^{\mathrm{eq}}$ are the thermal-equilibrium site fractions and $k_{\mathrm{V}}^{\mathrm{GB}}$, $k_{\mathrm{I}}^{\mathrm{GB}}$ are the grain-boundary sink efficiencies of the two defects. The perfect-sink (Dirichlet) condition $C_{\mathrm{V}} = C_{\mathrm{V}}^{\mathrm{eq}}$, $C_{\mathrm{I}} = C_{\mathrm{I}}^{\mathrm{eq}}$ is recovered for $k^{\mathrm{GB}} \to \infty$. With the Dirichlet condition the defect gradients at the boundary, and with them both the magnitude and the width of the segregation profiles, do not converge under mesh refinement, a behaviour also reported for other Onsager-type RIS models [31]; the finite-efficiency condition removes this dependence (Supplementary Information, Sec. S1). The sink efficiencies are calibrated together with the defect-mediated transport coefficients, Sec. 2.4.1.

2.2 Grain-Boundary Oxidation Model

The oxidized zone is treated as a channel of half-width $L(y,t)$, the distance from the boundary mid-plane to the oxide/metal interface. The unoxidized boundary is an open channel of half-width δ_{GB}, and the oxide of total thickness $S = \sum_k L_k$ grows from the interface into the metal, so that

$$L = \max\left(\delta_{\mathrm{GB}},\, S\right); \tag{15}$$

the part of the section not yet filled, $L - S$, remains open channel with the oxygen diffusivity D_{O}^{0} of the unoxidized boundary.

The intergranular oxide formed on 304NG in PWR primary water gives an amorphous-like fast Fourier transform in high-resolution TEM [11], and the oxide core along the boundary plane of proton-irradiated 316L is amorphous and porous [27]; the oxide is therefore not treated as an assembly of crystalline phases. The three oxides Cr_2O_3, Fe_3O_4 and SiO_2 are the end-members of a locally mixed oxide: they carry the stoichiometry of the Cr, Fe and Si consumption and the oxygen diffusivity assigned to each component, and their volume fractions f_k describe the local composition of the mixed oxide rather than the amounts of three distinct phases.

The composition of the oxide is not uniform across the section. New oxide forms at the interface $x = S$ with the composition of the metal consumed there at that time, the fractions $f_k^{\mathrm{new}} = \dot{L}_k/\dot{S}$ of the thickness rates of the three end-members, Eqs. (30)–(32), and is not modified once formed. The position x across the section is therefore a record of the time of formation, with the oldest oxide at the mid-plane and the newest at the interface, on both sides of the boundary. The local diffusivity at position x is

$$D_{\mathrm{O}}(x) = \sum_{k\in\{\mathrm{Cr_2O_3},\mathrm{Fe_3O_4}\}} f_k(x)\, D_{\mathrm{O}}^{k} + D_{\mathrm{O}}^{\mathrm{por}} \left[f_{\mathrm{SiO_2}}(x)\right]^{m}, \tag{16}$$

where D_{O}^{k} is the oxygen diffusivity of end-member k and $D_{\mathrm{O}}^{\mathrm{por}}$ that of the porosity left by the dissolution of SiO_2, whose volume fraction is $f_{\mathrm{SiO_2}}$. The two oxides enter as a volume-fraction average. The porosity enters through Archie's law, in which the transport coefficient of a porous medium scales with the porosity raised to a power m [6, 24]: the pores left by the dissolution connect to one another only in proportion to their volume fraction, and $m = 2$, within the range 1.9–2.0 derived for three-dimensional porous media from continuum percolation [6, 24], is held fixed. For $m = 1$ the porosity would be a parallel path and Eq. (16) would reduce to the plain volume-fraction average.

Oxygen moves along the boundary through the whole section in parallel and reaches the interface across it in series, so an oxide whose diffusivity varies across the section has two effective diffusivities,

$$D_{\mathrm{O}}^{\mathrm{eff},y} = \frac{1}{L}\left[\int_0^S D_{\mathrm{O}}(x)\,\mathrm{d}x + D_{\mathrm{O}}^{0}\,(L-S)\right], \qquad D_{\mathrm{O}}^{\mathrm{eff},x} = L\left[\int_0^S \frac{\mathrm{d}x}{D_{\mathrm{O}}(x)} + \frac{L-S}{D_{\mathrm{O}}^{0}}\right]^{-1}, \tag{17}$$

the arithmetic and harmonic means that are the exact transport coefficients of a medium whose diffusivity varies in one direction, for flux parallel and normal to that direction, and that form the upper and lower Wiener bounds of effective-medium theory [8, 22]. The two integrals are advanced in time together with the oxide,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^S D_{\mathrm{O}}\,\mathrm{d}x = D_{\mathrm{O}}^{\mathrm{new}}\,\dot{S}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\int_0^S \frac{\mathrm{d}x}{D_{\mathrm{O}}} = \frac{\dot{S}}{D_{\mathrm{O}}^{\mathrm{new}}}, \tag{18}$$

where $D_{\mathrm{O}}^{\mathrm{new}}$ is Eq. (16) evaluated for the oxide forming at the interface. The composition at every position thus acts on the transport without the profile $f_k(x)$ being stored; the composition profiles across the oxide reported in Sec. 3 are reconstructed from the same growth record.

Integrating the two-dimensional oxygen balance over the cross-section gives, for the section-averaged concentration $\bar{C}_{\mathrm{O}}(y,t)$,

$$\frac{\partial}{\partial t}\big(L\,\bar{C}_{\mathrm{O}}\big) = -\frac{\partial}{\partial y}\big(L\,J_{\mathrm{O}}\big) - J_{\mathrm{r}}, \tag{19}$$

$$J_{\mathrm{O}} = -D_{\mathrm{O}}^{\mathrm{eff},y}\,\frac{\partial \bar{C}_{\mathrm{O}}}{\partial y}, \tag{20}$$

where J_{r} is the oxygen flux reaching the interface. Equation (19) is exact up to terms of order $\partial L/\partial y$ and makes no assumption on the lateral concentra-

tion profile; L inside the time derivative accounts for the dilution of dissolved oxygen as the section widens (Supplementary Information, Sec. S3).

The relation between J_r and $\bar{C}_\mathrm{O}$ is obtained from the fundamental separable mode of the steady diffusion equation in the cross-section, $C_\mathrm{O} \propto \cos(\mu x/L)$, which satisfies the symmetry condition at the mid-plane and the reaction condition at the interface, with the across-section diffusivity $D_\mathrm{O}^{\mathrm{eff},x}$ of Eq. (17). It gives

$$J_\mathrm{r} = \frac{D_\mathrm{O}^{\mathrm{eff},x}\,\mu^2\,\bar{C}_\mathrm{O}}{L}, \qquad C_\mathrm{O}^\mathrm{int} = \bar{C}_\mathrm{O}\,\mu\cot\mu, \tag{21}$$

where $C_\mathrm{O}^\mathrm{int}$ is the oxygen concentration at the interface and the dimensionless eigenvalue $\mu \in (0, \pi/2)$ is fixed at every position by the interfacial balance between oxygen supply and consumption,

$$\frac{D_\mathrm{O}^{\mathrm{eff},x}\,\mu^2\,\bar{C}_\mathrm{O}}{L} = \sum_k J_\mathrm{r}^k\big(\bar{C}_\mathrm{O}\,\mu\cot\mu\big)\,. \tag{22}$$

The interfacial reaction rates of the three end-members are

$$J_\mathrm{r}^{\mathrm{Cr_2O_3}} = k_\mathrm{r}^{\mathrm{Cr_2O_3}}\,C_\mathrm{O}^\mathrm{int}\,C_\mathrm{Cr}^\mathrm{GB}, \tag{23}$$

$$J_\mathrm{r}^{\mathrm{Fe_3O_4}} = k_\mathrm{r}^{\mathrm{Fe_3O_4}}\,C_\mathrm{O}^\mathrm{int}\,C_\mathrm{Fe}^\mathrm{GB}, \tag{24}$$

$$J_\mathrm{r}^{\mathrm{SiO_2}} = k_\mathrm{r}^{\mathrm{SiO_2}}\,C_\mathrm{O}^\mathrm{int}\,C_\mathrm{Si}^\mathrm{GB}, \tag{25}$$

where k_r^k is the interfacial rate constant of end-member k; their sum is the total interfacial consumption,

$$J_\mathrm{r} = J_\mathrm{r}^{\mathrm{Cr_2O_3}} + J_\mathrm{r}^{\mathrm{Fe_3O_4}} + J_\mathrm{r}^{\mathrm{SiO_2}}. \tag{26}$$

The left-hand side of Eq. (22) increases and the right-hand side decreases monotonically with μ, so the root is unique; it is found by Newton's method. In the reaction-limited limit, $\mu^2 \simeq kL/D_\mathrm{O}^{\mathrm{eff},x}$ with $k = \mathrm{d}J_\mathrm{r}/\mathrm{d}C_\mathrm{O}^\mathrm{int}$, the interface sees the average concentration, $C_\mathrm{O}^\mathrm{int} \simeq \bar{C}_\mathrm{O}$; in the diffusion-limited limit $\mu \to \pi/2$, $C_\mathrm{O}^\mathrm{int} \to 0$ and $J_\mathrm{r} \to \pi^2 D_\mathrm{O}^{\mathrm{eff},x}\bar{C}_\mathrm{O}/4L$. The closure thus spans both regimes without a prescribed transport resistance; its derivation, the neglected higher modes and its range of validity are given in the Supplementary Information, Sec. S3.

The oxygen concentration is normalized by the reference oxygen density $C_\mathrm{O}^\mathrm{ref}$ of the water-filled channel, so that the fluxes J_r^k have the dimension of

a velocity, whereas the metal concentrations are site fractions on a lattice of site density N. The rates of metal removal, in the same units as the metal fluxes, follow from the stoichiometry of each end-member,

$$J_{\mathrm{r}}^{\mathrm{Cr}} = \tfrac{2}{3}\,\frac{C_{\mathrm{O}}^{\mathrm{ref}}}{N}\,J_{\mathrm{r}}^{\mathrm{Cr_2O_3}}, \tag{27}$$

$$J_{\mathrm{r}}^{\mathrm{Fe}} = \tfrac{3}{4}\,\frac{C_{\mathrm{O}}^{\mathrm{ref}}}{N}\,J_{\mathrm{r}}^{\mathrm{Fe_3O_4}}, \tag{28}$$

$$J_{\mathrm{r}}^{\mathrm{Si}} = \tfrac{1}{2}\,\frac{C_{\mathrm{O}}^{\mathrm{ref}}}{N}\,J_{\mathrm{r}}^{\mathrm{SiO_2}}, \tag{29}$$

and the thickness of each end-member grows at the rate at which its formula units are produced,

$$\dot{L}_{\mathrm{Cr_2O_3}} = \tfrac{1}{3}\,\Omega_{\mathrm{Cr_2O_3}}\,C_{\mathrm{O}}^{\mathrm{ref}}\,J_{\mathrm{r}}^{\mathrm{Cr_2O_3}}, \tag{30}$$

$$\dot{L}_{\mathrm{Fe_3O_4}} = \tfrac{1}{4}\,\Omega_{\mathrm{Fe_3O_4}}\,C_{\mathrm{O}}^{\mathrm{ref}}\,J_{\mathrm{r}}^{\mathrm{Fe_3O_4}}, \tag{31}$$

$$\dot{L}_{\mathrm{SiO_2}} = \tfrac{1}{2}\,\Omega_{\mathrm{SiO_2}}\,C_{\mathrm{O}}^{\mathrm{ref}}\,J_{\mathrm{r}}^{\mathrm{SiO_2}}, \tag{32}$$

where $\Omega_k = M_k/(N_{\mathrm{A}}\rho_k)$ is the volume per formula unit of end-member k, with M_k its molar mass and ρ_k its density, and the prefactors are the reciprocals of the number of oxygen atoms per formula unit. SiO_2 is treated as dissolved on formation, so $L_{\mathrm{SiO_2}}$ is the thickness its volume would occupy, which is retained as porosity in Eq. (16). The total thickness $S = \sum_k L_k$ and the composition f_k^{new} of the oxide forming at the interface, of Sec. 2.2 follow from Eqs. (30)–(32).

Interfacial mass redistribution. The element with the largest ratio of consumption to availability sets the rate at which the oxidation front advances into the metal,

$$v_{\mathrm{lat}}^{\mathrm{ox,max}} = \max_i \frac{J_{\mathrm{r}}^{i}}{C_i^{\mathrm{GB}}}, \tag{33}$$

with $J_{\mathrm{r}}^{\mathrm{Ni}} = 0$ since Ni is not oxidized. The front sweeps element i at the rate $v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}}$, of which only J_{r}^{i} is oxidized; the excess

$$E_i = \max\!\big(v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}} - J_{\mathrm{r}}^{i},\, 0\big) \tag{34}$$

is either rejected into the metal ahead of the front or carried into the oxide, where it is retained as unoxidized solute or transported along the oxide grain

boundaries to the surface [10]. The partition between the two paths is treated as a nonequilibrium solute capture at the moving oxide/metal interface [36, 43], the oxidation analogue of solute trapping during rapid solidification [4, 5]: an element escapes the advancing front only by diffusive exchange across the interface, so the fraction incorporated is set by the ratio of the front velocity to a diffusive velocity. In the continuous-growth model of Aziz [4] the partition coefficient of the growing phase is $k(v) = (v/v_D + k_e)/(v/v_D + 1)$, with v_D the diffusive velocity of the interface and k_e the equilibrium partition coefficient. The excess E_i is metal that the oxide does not accommodate at equilibrium, $k_e = 0$, so the fraction carried into the oxide is

$$\lambda_i = \frac{v_{\mathrm{lat}}^{\mathrm{ox,max}}}{v_{\mathrm{lat}}^{\mathrm{ox,max}} + v_i^{\mathrm{r}}}, \tag{35}$$

where the redistribution velocity v_i^{r} takes the place of the diffusive velocity of element i, the characteristic velocity at which it can be rejected into the metal ahead of the front. Complete rejection ($\lambda_i = 0$), the local-equilibrium limit of classical oxidation theory [36], is recovered for $v_i^{\mathrm{r}} \to \infty$; for $v_i^{\mathrm{r}} \ll v_{\mathrm{lat}}^{\mathrm{ox,max}}$ the front outruns the rejection and the excess is carried into the oxide ($\lambda_i \to 1$), the capture regime that Yu et al. [43] argue is the common case in aqueous oxidation. The net removal of element i from the metal is then

$$\tilde{J}_{\mathrm{r}}^{i} = J_{\mathrm{r}}^{i} + \lambda_i E_i, \tag{36}$$

which replaces J_{r}^{i} in the interfacial conditions of Sec. 2.3.1. The motivation for this treatment and its limiting behaviour are given in the Supplementary Information, Sec. S2.

2.3 Coupling Strategy and Simulation Protocol

2.3.1 Lattice Velocity and Metal Consumption

Metal atoms leave the metal domain only at the grain-boundary plane, either by oxidation or by removal through the oxide, Eq. (36). Because the metal lattice is incompressible, this removal creates no source in the interior: the entire lattice translates towards the interface at the velocity set by the net interfacial removal. Denoting by

$$v_{\mathrm{ox}}(y,t) = \sum_i \tilde{J}_{\mathrm{r}}^{i} = \sum_i \left(J_{\mathrm{r}}^{i} + \lambda_i E_i \right) \tag{37}$$

the recession rate of the metal/oxide interface, which lies between the oxidation rate $\sum_i J_{\mathrm{r}}^i$ (complete rejection) and the sweep velocity $v_{\mathrm{lat}}^{\mathrm{ox,max}}$ (complete removal), the total lattice velocity in the metal domain is

$$\mathbf{v}_{\mathrm{lat}}(x,y,t) = \mathbf{v}_{\mathrm{lat}}^{\mathrm{RIS}}(x,y,t) + v_{\mathrm{ox}}(y,t)\,\mathbf{n}. \tag{38}$$

The two contributions differ in character. The segregation term is local: it is determined pointwise by the defect fluxes at (x, y). The oxidation term is not; it is evaluated at the boundary and is independent of x, so a single value applies across the whole domain at a given y. Segregation therefore redistributes the lattice internally, whereas oxidation translates it rigidly, so that the total flux of alloying element i entering Eq. (1) is

$$\mathbf{J}_i = \mathbf{J}_i^{\mathrm{V}} + \mathbf{J}_i^{\mathrm{I}} + C_i \mathbf{v}_{\mathrm{lat}}, \tag{39}$$

subject to the interfacial condition

$$\mathbf{J}_i \cdot \mathbf{n}\big|_{x=0} = \tilde{J}_{\mathrm{r}}^i, \tag{40}$$

which removes from the boundary plane the atoms incorporated into the growing oxide or carried away through it.

Equations (39) and (40) collect the three ways in which an alloying element can move. The first two terms are the diffusive fluxes carried by the two defect mechanisms, Eqs. (4) and (5); the third is the drift of the entire lattice relative to the laboratory frame; the interfacial consumption enters not as a bulk term but as an outgoing flux on the boundary plane.

The two contributions to $\mathbf{v}_{\mathrm{lat}}$ have different origins but act identically on every species. $\mathbf{v}_{\mathrm{lat}}^{\mathrm{RIS}}$ arises because the vacancy and interstitial fluxes are unequal, so lattice sites are created and destroyed at different rates; v_{ox} arises because oxidation consumes metal atoms from the interfacial layer, so the lattice is displaced towards the advancing front. Because the drift term $C_i \mathbf{v}_{\mathrm{lat}}$ is proportional to the local concentration and carries no species selectivity, it does not by itself alter the relative composition; selectivity enters only through the mechanism-specific diffusivities in Eqs. (6)–(9), through the stoichiometric sinks in Eqs. (27)–(29), and through the element-specific redistribution fractions λ_i of Eq. (35).

2.3.2 Sequential Irradiation–Oxidation Protocol

The reference experiments irradiate the specimen first and expose it to the primary water environment afterwards. The model reproduces this protocol

as two consecutive stages that share the same governing equations and differ only in two switches: the dose rate K_0 and the oxygen boundary condition at the grain-boundary mouth.

In the irradiation stage the specimen is irradiated to the target dose with no oxygen present, $K_0 > 0$ and $C_{\mathrm{O}} \equiv 0$, so that Eqs. (1)–(12) govern the evolution alone and all interfacial reaction rates vanish identically. The stage is integrated to $t_{\mathrm{irr}} = \Phi/K_0$, where Φ is the nominal dose in dpa, and yields the segregated composition profiles $C_i(x, y, t_{\mathrm{irr}})$.

The oxidation stage continues from the final state of the irradiation stage: all concentration fields, including the point-defect populations, are inherited unchanged,

$$(C_i,\, C_{\mathrm{V}},\, C_{\mathrm{I}})\big|_{t=0^+} = (C_i,\, C_{\mathrm{V}},\, C_{\mathrm{I}})\big|_{t=t_{\mathrm{irr}}}\,, \qquad L_k\big|_{t=0^+} = 0, \tag{41}$$

and the exposure clock is restarted from zero. The dose rate is then set to zero, $K_0 = 0$, so that the inherited defect supersaturation decays by recombination according to Eqs. (2) and (3) without further production.

Simultaneously, the oxygen boundary condition at the mouth of the boundary is switched on. Oxygen ingress is limited by the surface film that grows on the free surface, whose resistance increases as the film thickens; this is imposed as a Robin condition with a time-decaying transfer coefficient,

$$J_{\mathrm{O}}\big|_{y=0} = h(t)\left(C_{\mathrm{O}}^{\mathrm{w}} - \bar{C}_{\mathrm{O}}\big|_{y=0}\right), \qquad h(t) = \frac{k_{\mathrm{s}}}{\sqrt{t} + \sqrt{t_0}}, \tag{42}$$

where $C_{\mathrm{O}}^{\mathrm{w}}$ is the oxygen concentration of the bulk water, k_{s} is the surface transfer coefficient, and t_0 regularizes the initial instant. The $t^{-1/2}$ decay reproduces the parabolic thickening of the surface film, so that the cumulative oxygen admitted through the mouth grows as $t^{1/2}$. The deep end of the boundary is closed, $J_{\mathrm{O}}\big|_{y=L_{\mathrm{GB}}} = 0$.

Because no point defects are produced once $K_0 = 0$, the inherited supersaturation decays on the recombination timescale, which is short compared with the exposure time; segregation therefore effectively ceases early in the exposure stage, and the composition field that the oxidation model sees is the one established during irradiation. The sequential protocol thus isolates the compositional pathway: any dose dependence of the predicted oxidation depth must be carried by the radiation-induced segregation profile inherited through Eq. (41), since neither radiolysis nor sustained irradiation-assisted transport is present.

2.3.3 Fully Coupled Irradiation–Oxidation Framework

The sequential protocol of Sec. 2.3.2 is a special case of the governing equations rather than a separate model: it is obtained by switching K_0 and the mouth boundary condition on and off in turn. Leaving both switches on throughout, $K_0 > 0$ together with the Robin condition Eq. (42), gives the fully coupled problem in which segregation and oxidation proceed simultaneously, as they do in reactor service. No additional equations or parameters are required; Eqs. (1)–(12) are integrated as a single stage.

In this mode the two processes feed back on each other. Segregation continues to modify the boundary composition while the oxidation front advances, so the composition that the interfacial reactions see evolves in time rather than being fixed at its end-of-irradiation value; conversely, the metal consumed at the front and the associated lattice drift, Eq. (37), alter the concentration field on which the segregation fluxes act. This mode takes no part in the calibration of Sec. 2.4 and is used only for prediction.

In-service calculation. The fully coupled mode is applied to a component that was irradiated and oxidized together in service: a type 316 stainless steel baffle-former bolt removed from a Westinghouse two-loop PWR after about 40 years, with a dose of about 41 dpa at the bolt head [20, 28, 29]. On the water-exposed head-to-shank transition of this bolt, Gussev et al. [20] distinguished two kinds of intergranular damage: shallow, wedge-shaped oxidized grain boundaries, mostly 2 to 3 µm deep, and deeper short cracks of 4 µm or more. The shallow damage is the quantity compared with the calculated penetration depth; the cracks involve stress and are outside the model.

The calculation differs from the calibrated one only in its inputs. The bulk site fractions are those of the bolt, taken from the atom-probe composition reported by Lach et al. [28]: 0.1932 Cr, 0.1155 Ni and 0.0124 Si, with Fe as the balance, 0.6789, which includes Mo, Mn and the minor elements as in Table 1. The flux history of the bolt is not available [20], so the dose rate is taken as constant in time, $K_0 = 3.17 \times 10^{-8}\,\mathrm{dpa\,s^{-1}}$, that is 1 dpa per year and 40 dpa after the 40 years (350 400 h) of the calculation. Irradiation and oxygen ingress start together at $t = 0$ from the unsegregated alloy and continue to the end. All other parameters, those of Tables 2–4, keep their calibrated values; none is adjusted to the in-service data. The domain is extended to 5 µm along the boundary, with a mesh spacing of 50 nm in that

direction and 1 nm normal to the boundary, and the penetration depth is defined as in Sec. 2.4.3.

2.4 Model Parameterization and Calibration

The two components are parameterized and calibrated separately and against separate data. The RIS model is calibrated first; the oxidation model is then calibrated with the RIS model related parameters fixed. The bulk composition of the alloy, which sets the initial and far-field site fractions of the four elements in both components, is given in Table 1.

Table 1: Composition of the 304NG stainless steel of the reference experiments [11] and the bulk site fractions used in the model as initial and far-field values of the four elements. The site fractions are the atomic fractions of the substitutional lattice: C, an interstitial solute, is excluded, and Mn, Co, P and S are counted with Fe.

	Fe	Cr	Ni	Si	Mn	C	P	S	Co
Measured (wt%)	bal.	19.51	8.87	0.27	1.73	0.04	0.021	0.002	0.04
Atomic (at.%)	68.51	20.65	8.32	0.53	1.73	0.18	0.04	0.003	0.04
Model site fraction	0.7045	0.2069	0.0833	0.0053	–	–	–	–	–

2.4.1 RIS Parameterization

The RIS model requires the mechanism-specific diffusion coefficients D_i^{V} and D_i^{I} of the four alloying elements, the correlation factors f_0^{V} and f_0^{I}, the recombination coefficient R, the damage efficiency ϵ, the dose rate K_0, and the grain-boundary sink efficiencies $k_{\mathrm{V}}^{\mathrm{GB}}$ and $k_{\mathrm{I}}^{\mathrm{GB}}$ of Eqs. (13) and (14). The dose rate and irradiation temperature are those of the reference experiment. The correlation factors are taken from their crystallographic values for the fcc lattice. The damage efficiency accounts for the fraction of displacements that survive in-cascade recombination and escape as freely migrating defects; its value is that reported for proton irradiation. The mechanism-specific diffusion coefficients and the grain-boundary sink efficiencies are calibrated against the measured grain-boundary composition profiles of the reference experiment [9], as shown in Figure 3. The interstitial-mediated diffusion coefficients D_i^{I} are effective values. In the multicomponent rate-theory

model of FUKUYA and FUJII [18], the preferential association of undersized solutes with self-interstitials is represented by a coupling parameter $\xi_k = \exp(E^{\mathrm{b}}_{k\mathrm{I}}/k_B T) \big/ \sum_m C_m \exp(E^{\mathrm{b}}_{m\mathrm{I}}/k_B T)$, the fraction of self-interstitials that form a mixed dumbbell with solute k, set by the solute–interstitial binding energy $E^{\mathrm{b}}_{k\mathrm{I}}$, which is taken as nonzero for Si alone. In the rate equations this parameter appears only as the product $\xi_k d^{\mathrm{I}}_k$ with the interstitial partial diffusion coefficient, so it is absorbed here into D^{I}_i: the strong binding of Si to self-interstitials appears as a larger effective $D^{\mathrm{I}}_{\mathrm{Si}}$, and the weak dependence of ξ_k on the local composition through its normalization is neglected. Related parameters are shown in Table 2.

2.4.2 Interfacial Reaction and Redistribution Parameterization

The interfacial rate constants k^k_{r} of the three end-members and the redistribution velocities v^{r}_i of Eq. (35) are fixed by the EDX line scan across the intergranular oxide of the unirradiated specimen of Deng et al. [11]: the metal-basis atomic fractions of the oxide and the Ni content of the metal at the oxide/metal interface (Fig. 4a–c). The scans of the irradiated specimens take no part in the calibration. Fe is soluble in the oxide and diffuses through the intergranular oxide along its grain boundaries to the surface, where it feeds the growth of the outer spinel particles [10]; it is therefore assumed not to be rejected into the metal, and $v^{\mathrm{r}}_{\mathrm{Fe}}$ is set to a value small compared with the front velocity, the limit $\lambda_{\mathrm{Fe}} \to 1$ in which the Fe removed is the amount swept at the metal-side composition, Eq. (34). The same value is used for Cr and Si, which set or nearly set the front velocity in Eq. (33), so that their excess is small. Ni is not oxidized and its rejection builds up the enrichment of the metal ahead of the interface; $v^{\mathrm{r}}_{\mathrm{Ni}}$ is calibrated against the Ni content of the metal at the interface of the unirradiated specimen after oxidation. The rate constants $k^{\mathrm{Cr_2O_3}}_{\mathrm{r}}$ and $k^{\mathrm{Fe_3O_4}}_{\mathrm{r}}$ are calibrated against the Cr and Fe fractions of the unirradiated oxide. The oxide composition does not constrain $k^{\mathrm{SiO_2}}_{\mathrm{r}}$, since the SiO_2 formed is treated as dissolved; it is set above $k^{\mathrm{Cr_2O_3}}_{\mathrm{r}}$ and held fixed, under the constraint $k^{\mathrm{SiO_2}}_{\mathrm{r}} > k^{\mathrm{Cr_2O_3}}_{\mathrm{r}} \gg k^{\mathrm{Fe_3O_4}}_{\mathrm{r}}$ imposed by the order of thermodynamic stability of the three oxides, SiO_2 being the most and Fe_3O_4 the least stable [42]. The reference oxygen density and lattice site density of Eqs. (27)–(29) and the oxide densities of Eqs. (30)–(32) are fixed. The values are given in Table 3; they are held fixed in the calibration of the transport parameters that follows.

2.4.3 Oxygen Transport Parameterization

With the interfacial parameters fixed, the oxygen diffusivities $D_{\mathrm{O}}^{\mathrm{Cr_2O_3}}$ and $D_{\mathrm{O}}^{\mathrm{Fe_3O_4}}$ of the two oxide end-members and the porosity diffusivity $D_{\mathrm{O}}^{\mathrm{por}}$ of Eq. (16) are calibrated against the dose-dependent grain-boundary oxidation depths of Deng et al. [11], measured after exposure in simulated PWR primary water at 0, 0.5, and 3.0 dpa. The predicted oxidation depth is defined as the distance from the boundary mouth to the deepest position at which the total oxide thickness S exceeds the channel half-width δ_{GB}, applied identically at every dose. The three diffusivities are adjusted to reproduce the measured depths at all three doses simultaneously, using a single parameter set; no parameter is allowed to take different values at different doses. The remaining transport parameters are not adjusted: the Archie exponent is held at $m = 2$, the open-channel diffusivity D_{O}^{0} is set equal to the porosity diffusivity, the channel half-width δ_{GB} is assigned, and the surface transfer coefficient k_{s} and regularization time t_0 of Eq. (42) are assigned fixed values, to which the predicted depths are insensitive. The values are given in Table 4.

The two levels of calibration are not fully independent. The oxide composition fixes the ratios of the rate constants, whereas the total oxide thickness depends on the rate constants and the diffusivities together through the closure of Eq. (22), which partitions the oxygen supply between interfacial reaction and transport across the oxide. The absolute level of the rate constants is therefore adjusted, together with the diffusivities, so that the calculated total thickness of the intergranular oxide is close to the measured one; it is not fitted more precisely.

3 Results

3.1 Radiation-Induced Segregation Profiles

Figure 3 compares the calculated and measured grain-boundary composition profiles after irradiation to 0.5 and 3.0 dpa. The calculated profiles are convolved with a 1 nm Gaussian to account for the finite spatial resolution of the measurements. These profiles are the target of the RIS calibration of Sec. 2.4.1.

Cr is depleted at the boundary and Ni and Si are enriched, at both doses. The calculated boundary Cr concentration falls from the matrix value of

Table 2: Parameters of the radiation-induced segregation model. The mechanism-specific diffusion coefficients multiply the defect site fraction in the Onsager coefficients, Eqs. (6)–(9); $D^{\mathrm{I}}_{\mathrm{Si}}$ is the effective value that absorbs the Si–interstitial coupling parameter (Sec. 2.4.1).

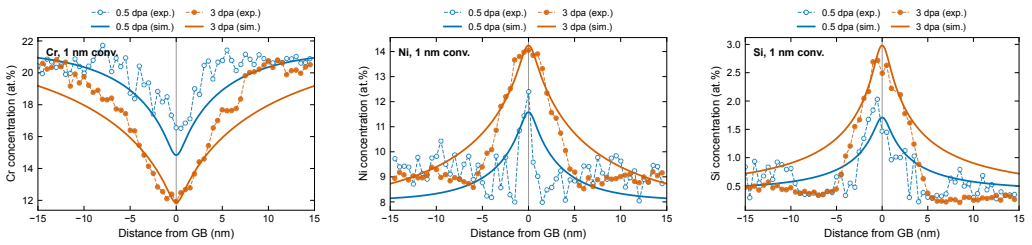
Symbol	Description	Value	Source
K_0	Dose rate	$6 \times 10^{-6}\,\mathrm{dpa\,s^{-1}}$	Deng et al. [11]
	Dose (irradiation time = dose/K_0)	0.5, 3.0 dpa	Deng et al. [11]
ϵ	Damage efficiency	0.2	Protons [2]
R	Recombination coefficient	$1 \times 10^{4}\,\mathrm{s^{-1}}$	Calibrated against the grain-boundary composition profiles
$C^{\mathrm{eq}}_{\mathrm{V}}$	Equilibrium vacancy site fraction	1×10^{-13}	Assumed
$C^{\mathrm{eq}}_{\mathrm{I}}$	Equilibrium interstitial site fraction	1×10^{-30}	Assumed
$D^{\mathrm{V}}_{\mathrm{Cr}}$	Vacancy-mediated diffusivity, Cr	$5.0 \times 10^{4}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the grain-boundary composition profiles
$D^{\mathrm{V}}_{\mathrm{Fe}}$	Vacancy-mediated diffusivity, Fe	$3.5 \times 10^{4}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the grain-boundary composition profiles
$D^{\mathrm{V}}_{\mathrm{Ni}}$	Vacancy-mediated diffusivity, Ni	$2.4 \times 10^{4}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the grain-boundary composition profiles
$D^{\mathrm{V}}_{\mathrm{Si}}$	Vacancy-mediated diffusivity, Si	$5.0 \times 10^{4}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the grain-boundary composition profiles
$D^{\mathrm{I}}_{\mathrm{Cr}}$	Interstitial-mediated diffusivity, Cr	$3.0 \times 10^{5}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the grain-boundary composition profiles
$D^{\mathrm{I}}_{\mathrm{Fe}}$	Interstitial-mediated diffusivity, Fe	$3.0 \times 10^{5}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the grain-boundary composition profiles
$D^{\mathrm{I}}_{\mathrm{Ni}}$	Interstitial-mediated diffusivity, Ni	$3.0 \times 10^{5}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the grain-boundary composition profiles
$D^{\mathrm{I}}_{\mathrm{Si}}$	Interstitial-mediated diffusivity, Si (effective, includes ξ_{Si})	$1.25 \times 10^{6}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the grain-boundary composition profiles
f^{V}_0	Vacancy correlation factor	0.78	fcc lattice
f^{I}_0	Interstitial correlation factor	0.44	fcc dumbbell [7]
$k^{\mathrm{GB}}_{\mathrm{V}}$	Grain-boundary sink efficiency, vacancies, Eq. (13)	$3 \times 10^{4}\,\mathrm{nm\,s^{-1}}$	Calibrated against the grain-boundary composition profiles
$k^{\mathrm{GB}}_{\mathrm{I}}$	Grain-boundary sink efficiency, interstitials, Eq. (14)	$2 \times 10^{5}\,\mathrm{nm\,s^{-1}}$	Calibrated against the grain-boundary composition profiles

Table 3: Parameters of the interfacial reaction and redistribution model, fixed by the composition of the intergranular oxide and the Ni content of the metal at the oxide/metal interface of the unirradiated specimen (Sec. 2.4.2). Oxygen concentrations are normalized by that of the bulk water, $C_{\mathrm{O}}^{\mathrm{w}} = 1$.

Symbol	Description	Value	Source
$k_{\mathrm{r}}^{\mathrm{Cr_2O_3}}$	Interfacial rate constant, Cr_2O_3, Eq. (23)	$6 \times 10^{-4}\,\mathrm{nm\,s^{-1}}$	Calibrated against the oxide composition, 0 dpa
$k_{\mathrm{r}}^{\mathrm{Fe_3O_4}}$	Interfacial rate constant, Fe_3O_4, Eq. (24)	$5 \times 10^{-5}\,\mathrm{nm\,s^{-1}}$	Calibrated against the oxide composition, 0 dpa
$k_{\mathrm{r}}^{\mathrm{SiO_2}}$	Interfacial rate constant, SiO_2, Eq. (25)	$7 \times 10^{-4}\,\mathrm{nm\,s^{-1}}$	Stability ordering, held fixed
$v_{\mathrm{Cr}}^{\mathrm{r}}$	Redistribution velocity, Cr, Eq. (35)	$1 \times 10^{-7}\,\mathrm{nm\,s^{-1}}$	Capture limit
$v_{\mathrm{Fe}}^{\mathrm{r}}$	Redistribution velocity, Fe, Eq. (35)	$1 \times 10^{-7}\,\mathrm{nm\,s^{-1}}$	Capture limit [10]
$v_{\mathrm{Si}}^{\mathrm{r}}$	Redistribution velocity, Si, Eq. (35)	$1 \times 10^{-7}\,\mathrm{nm\,s^{-1}}$	Capture limit
$v_{\mathrm{Ni}}^{\mathrm{r}}$	Redistribution velocity, Ni, Eq. (35)	$1.5 \times 10^{-5}\,\mathrm{nm\,s^{-1}}$	Calibrated against the interface Ni content, 0 dpa
$C_{\mathrm{O}}^{\mathrm{ref}}$	Reference oxygen density, Eqs. (27)–(32)	$22\,\mathrm{nm^{-3}}$	Assumed
N	Lattice site density of the fcc metal	$87\,\mathrm{nm^{-3}}$	fcc lattice, $a = 0.359\,\mathrm{nm}$
$\rho_{\mathrm{Cr_2O_3}}$	Density of Cr_2O_3, Eq. (30)	$5.22\,\mathrm{g\,cm^{-3}}$	Crystallographic
$\rho_{\mathrm{Fe_3O_4}}$	Density of Fe_3O_4, Eq. (31)	$5.17\,\mathrm{g\,cm^{-3}}$	Crystallographic
$\rho_{\mathrm{SiO_2}}$	Density of SiO_2, Eq. (32)	$2.2\,\mathrm{g\,cm^{-3}}$	Amorphous silica

Table 4: Parameters of the oxygen transport model and exposure conditions, with the three diffusivities fixed by the dose-dependent oxidation depths (Sec. 2.4.3).

Symbol	Description	Value	Source
t_{ox}	Exposure time	500 h	Deng et al. [11]
$C_{\mathrm{O}}^{\mathrm{w}}$	Oxygen concentration of the bulk water, Eq. (42)	1	Normalization
k_{s}	Surface transfer coefficient, Eq. (42)	$0.4\,\mathrm{nm\,s^{-1/2}}$	Assumed
t_0	Regularization time of Eq. (42)	100 s	Assumed
$D_{\mathrm{O}}^{\mathrm{Cr_2O_3}}$	Oxygen diffusivity in Cr_2O_3, Eq. (16)	$6 \times 10^{-5}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the oxidation depths
$D_{\mathrm{O}}^{\mathrm{Fe_3O_4}}$	Oxygen diffusivity in Fe_3O_4, Eq. (16)	$5 \times 10^{-3}\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the oxidation depths
$D_{\mathrm{O}}^{\mathrm{por}}$	Oxygen diffusivity of the porosity, Eq. (16)	$0.8\,\mathrm{nm^2\,s^{-1}}$	Calibrated against the oxidation depths
m	Archie exponent, Eq. (16)	2	Held fixed [24]
D_{O}^{0}	Oxygen diffusivity of the open boundary channel, Eq. (17)	$0.8\,\mathrm{nm^2\,s^{-1}}$	Set equal to $D_{\mathrm{O}}^{\mathrm{por}}$
δ_{GB}	Half-width of the unoxidized boundary, Eq. (15)	0.05 nm	Assumed



(a) (b) (c)

Figure 3: Calculated and measured grain-boundary composition profiles after irradiation to 0.5 and 3.0 dpa: (a) Cr, (b) Ni, (c) Si. Solid lines are calculated profiles obtained with the finite-efficiency sink condition, Eqs. (13) and (14), and convolved with a 1 nm FWHM Gaussian; symbols are the measurements of Deng et al. [9] at 0.5 and 3.0 dpa.

about 21 at.% to about 15 at.% at 0.5 dpa and about 12 at.% at 3.0 dpa, against measured minima of about 16.5 and 12 at.% (Fig. 3a). Ni rises from about 8 at.% to about 11.6 and 14.2 at.%, against measured maxima of about 12.5 and 14.5 at.% (Fig. 3b). Si rises from about 0.5 at.% to about 1.7 and 3.0 at.%, against measured maxima of about 2.0 and 2.7 at.% (Fig. 3c). The boundary values at 3.0 dpa are reproduced within 0.3 at.% for all three elements, with a calculated Cr-depleted zone somewhat wider than the measured one; at 0.5 dpa the model overestimates the Cr depletion by about 1.5 at.% and underestimates the Si enrichment by about 0.3 at.%.

3.2 Oxide Composition and Interfacial Chemistry

Figure 4 compares the calculated composition across the intergranular oxide with the EDX line scans of Deng et al. [11] after the 500 h exposure. The calculated profiles follow from the growth record of Sec. 2.2: the oxide formed first lies at the centre of the intergranular oxide and the later oxide grows outward to the oxide/metal interface on both sides.

The measurements show the signature of selective oxidation at all three doses: Cr is enriched and Fe depleted in the oxide relative to the matrix. The Cr content of the oxide, normalized to exclude oxygen, decreases with dose, from 74.9 at.% in the solution-annealed specimen to 69.5 and 58.3 at.% at 0.5 and 3.0 dpa [11]. On the metal side of the interface the scans show no significant Cr depletion or Fe enrichment, and only a slight Ni enrichment.

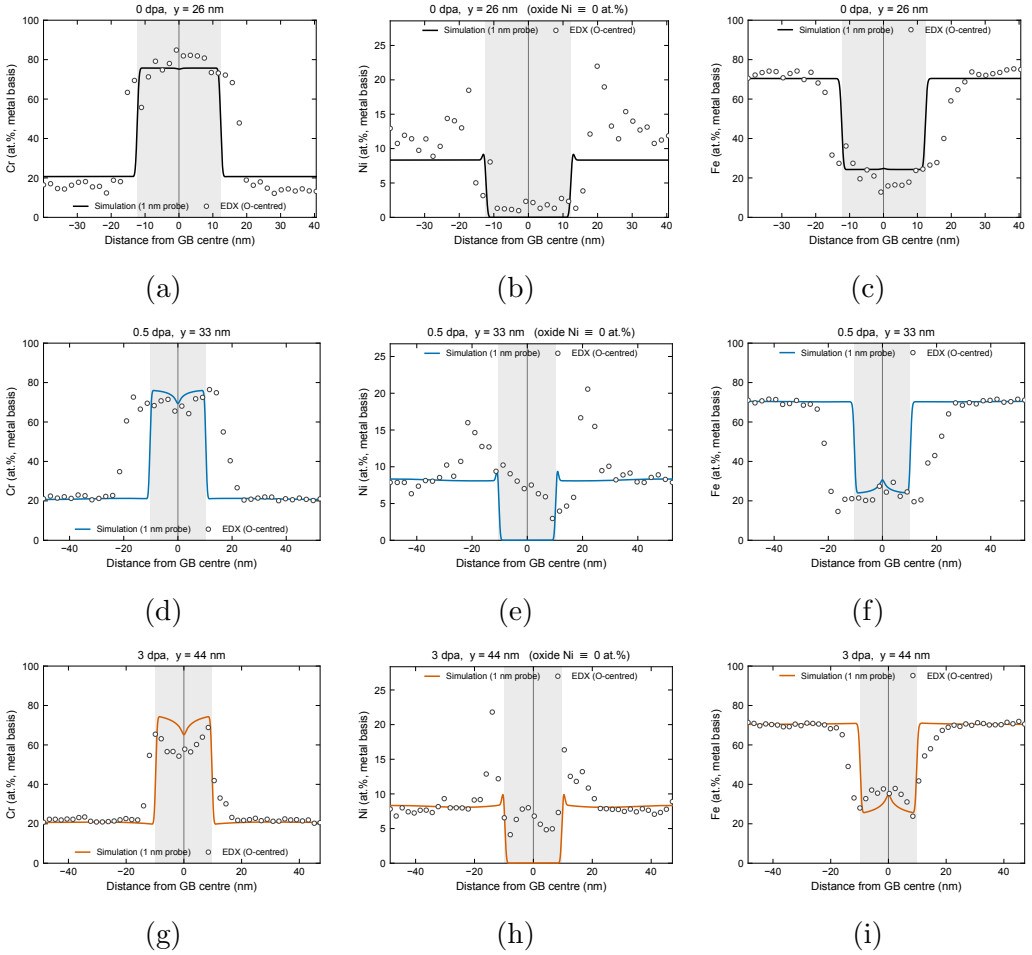
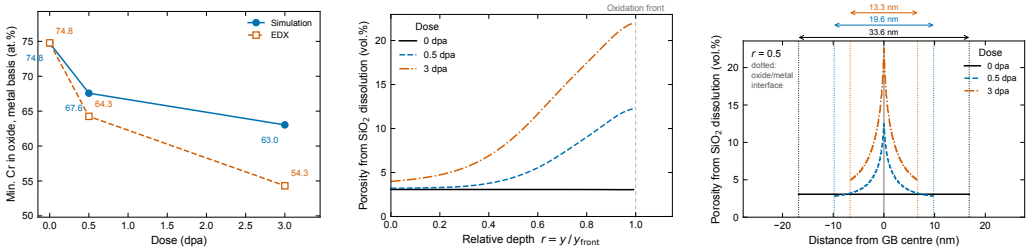


Figure 4: Calculated and measured composition profiles across the intergranular oxide after the 500 h exposure, at the depth of the EDX line scan: (a–c) solution-annealed, (d–f) 0.5 dpa, (g–i) 3.0 dpa; Cr, Ni and Fe from left to right. Shaded bands mark the calculated oxide. Calculated profiles are convolved with a Gaussian of 1 nm FWHM to match the EDX probe; measurements are the line scans of Deng et al. [11], centred on the oxygen signal.

In the irradiated specimens the Cr profile exhibits a minimum at the centre of the oxide that deepens with increasing dose.

The calculated profiles reproduce these features. With the interfacial mass redistribution of Eqs. (33)–(36), the calculated Cr profile retains the shape of the segregation profile across the interface, with neither Cr deple-

tion nor Fe enrichment on the metal side, and shows the central Cr minimum whose depth increases with dose; Fe shows the mirror image, a central maximum in a Fe-depleted oxide. The Cr content away from the centre stays close to its unirradiated value, because the outer part of the oxide forms from metal outside the segregated zone, which is only a few nanometres wide (Fig. 3). Ni is not oxidized in the model and the Ni swept by the front is carried out through the oxide (Eq. (35)), so the calculated Ni content of the oxide is zero, against measured values of 1 to 4 at.%; on the metal side the calculated Ni, convolved with the probe, rises only to 9 to 10 at.% immediately ahead of the interface at the depth of the scans, below the measured 12 to 16 at.%; at the boundary plane itself the calculated Ni after oxidation is 12.5, 13.4 and 14.5 at.% at 0, 0.5 and 3.0 dpa, against 20, 22 and 20 at.% reported by Deng et al. [11].



(a) (b) (c)

Figure 5: Cr content and SiO_2-dissolution porosity of the intergranular oxide after the 500 h exposure. (a) Calculated minimum Cr content of the intergranular oxide, on a metal basis, compared with the EDX values of Deng et al. [11]. (b) Porosity generated by SiO_2 dissolution along the boundary, averaged over the oxide section, as a function of relative depth, at 0, 0.5 and 3.0 dpa. (c) Porosity profile across the intergranular oxide at half the penetration depth; dotted lines mark the oxide/metal interfaces and the arrows give the oxide width.

The dose dependence of the oxide composition is shown in Fig. 5a. The calculated minimum Cr content at the centre of the oxide follows the measured decrease, from 74.8 at.% (calculated and measured) in the solution-annealed specimen to 67.6 and 64.3 at.% at 0.5 dpa and 63.0 and 54.3 at.% at 3.0 dpa (Fig. 5a).

With all SiO_2 treated as dissolved on formation, the porosity generated in

the intergranular oxide increases with dose and with depth along the boundary (Fig. 5b): averaged over the oxide section it rises from the unirradiated value of 3.1 vol% to about 12 and 22 vol% at the front at 0.5 and 3.0 dpa, while near the mouth, where the oxide is thickest, it stays at 3.2 and 4.0 vol%. Across the oxide section at half the penetration depth (Fig. 5c) the porosity is concentrated within a few nanometres of the boundary mid-plane, where the Si-enriched boundary was oxidized first, with peak values of about 12.5 and 22.5 vol% that fall toward the unirradiated value at the oxide/metal interface; the oxide at this depth is narrower at higher dose, 34, 20 and 13 nm at 0, 0.5 and 3.0 dpa, because the faster front leaves less time for it to thicken. A porous interface along the centre of the intergranular oxide, extending to the oxide tip, is what Kuang et al. [27] observed by under-focus TEM imaging of proton-irradiated 316L.

3.3 Intergranular Oxide Penetration Depth

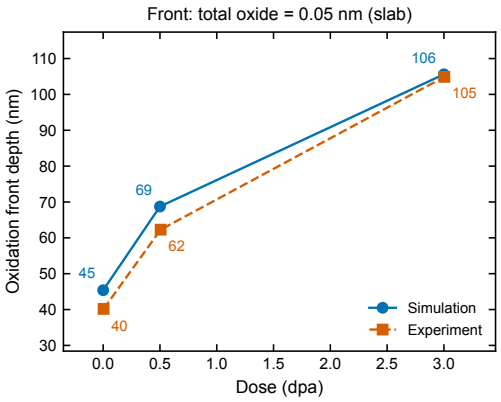
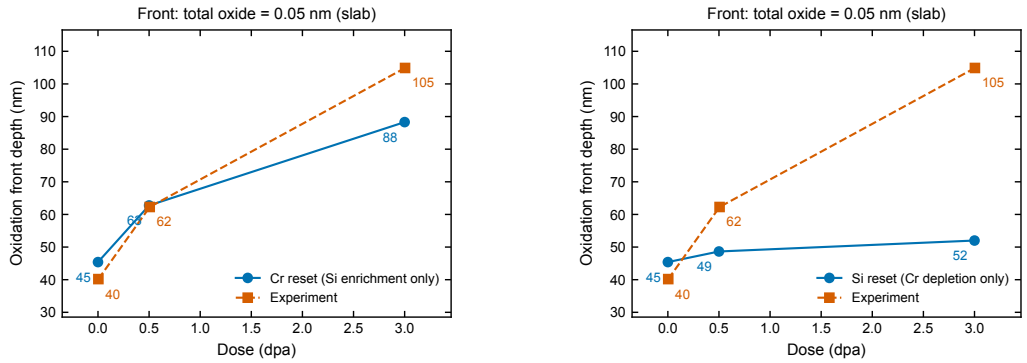


Figure 6: Intergranular oxide penetration depth after 500 h, calculated and measured [11]. These data calibrate the oxidation parameters.

The oxidation parameters are calibrated against the dose dependence of the intergranular oxide penetration depth after 500 h of exposure measured by Deng et al. [11] at 0, 0.5 and 3.0 dpa, Fig. 6. With a single parameter set the calculated depths are 45, 69 and 106 nm against measured values of 40, 62 and 105 nm: the depth at 3.0 dpa is matched within 1 nm, and the unirradiated and 0.5 dpa depths are overestimated by 5 and 7 nm, 13 and

10 %. The increase of penetration depth with dose is carried entirely by the segregation profiles inherited from the irradiation stage, Eq. (41).

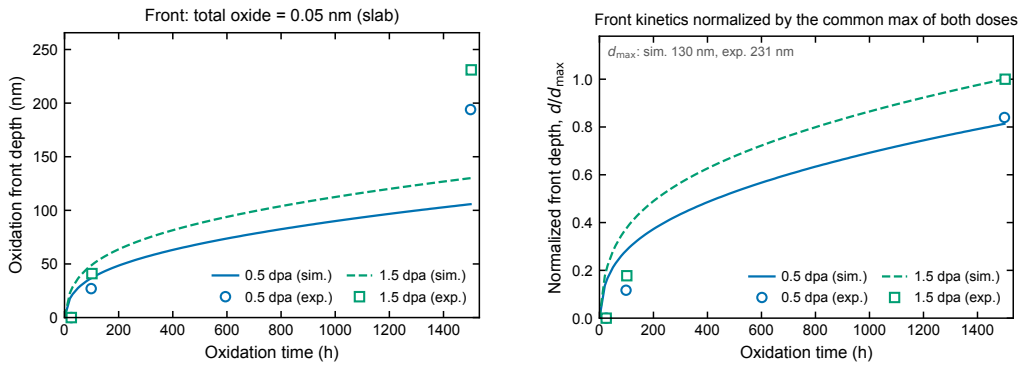


(a) (b)

Figure 7: Separation of the Cr and Si contributions to the dose dependence of the penetration depth after 500 h. (a) Cr segregation suppressed: the Cr profile is reset to its unirradiated value and the Si enrichment is retained. (b) Si segregation suppressed: the Si profile is reset and the Cr depletion is retained. All oxidation parameters are those of Fig. 6. Measurements from Deng et al. [11].

To separate the two routes by which segregation enters the oxidation kinetics, the calculation of Fig. 6 was repeated with one segregation profile at a time reset to its unirradiated value, all other inputs unchanged (Fig. 7). Both runs start from the unirradiated depth of 45 nm. With the Cr profile reset and only the Si enrichment retained, the depth increases to 63 and 88 nm at 0.5 and 3.0 dpa, against the measured 62 and 105 nm. With the Si profile reset and only the Cr depletion retained, the depth increases to 49 and 52 nm. Of the 60 nm increase of the full calculation between 0 and 3.0 dpa, the Si enrichment alone produces 43 nm and the Cr depletion alone 7 nm; the two routes are not additive, and their combination accounts for the remaining 11 nm.

With the parameters fixed by the calibration of Fig. 6, the model is used to predict the time dependence of the penetration depth, which is compared in Fig. 8 with the measurements of Deng et al. [10] at 0.5 and 1.5 dpa after 100 and 1500 h. After 100 h the calculated depths, 37 and 49 nm at 0.5 and 1.5 dpa, exceed the measured 27 and 41 nm. After 1500 h the measured



(a) (b)

Figure 8: Time dependence of the penetration depth at 0.5 and 1.5 dpa, predicted with the calibrated parameters held fixed, and measured by Deng et al. [10]: (a) absolute depth; (b) depths normalized by the maximum over both doses, $L/L_{\max}$, separately for calculations and measurements.

depths are 194 and 231 nm, whereas the model gives 106 and 130 nm (Fig. 8a). The two experimental series differ in their apparent kinetics: the apparent effective diffusivity $\bar{D}_{\text{eff}} = L^2/2t$ of the 0.5 dpa specimen of Deng et al. [10] after 1500 h exceeds that of the 3.0 dpa specimen of Deng et al. [11] after 500 h (Supplementary Information, Table S1). The measurements do not determine the time exponent of the penetration. The intergranular oxide was still flat after 25 h and had penetrated along the boundary after 100 h [10], so the penetration began at an unknown time between the two. With only two approximate depths per dose, the onset and the time exponent cannot be determined separately, and the time exponents of the calculation and the measurements are not compared; the comparison tests only the effect of dose, through the ratio of the penetration depths at the two doses. When the depths are normalized by the maximum over both doses, Fig. 8b, the calculated ratio between the two doses at 1500 h, 0.81, matches the measured 0.84; at that time the onset, at most 100 h, is a small part of the exposure and has little effect on the ratio. At 100 h the calculated normalized depths, 0.28 and 0.38, exceed the measured 0.12 and 0.18; this difference depends on the unknown onset and is not interpreted.

3.4 Fully Coupled Predictions under Concurrent Irradiation and Oxidation

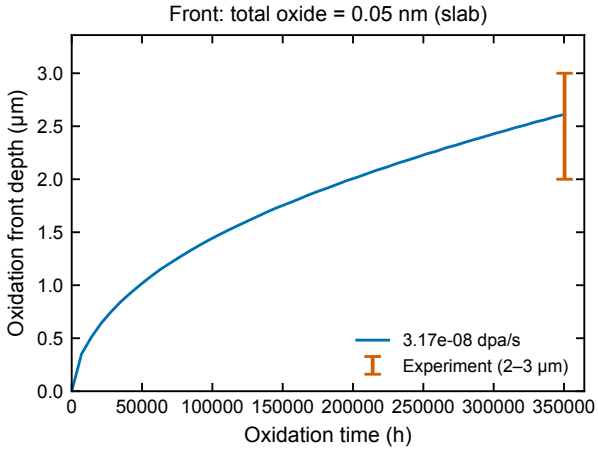


Figure 9: Penetration depth calculated for concurrent irradiation and oxidation of the type 316 baffle-former bolt at a constant dose rate of $3.17 \times 10^{-8}\,\mathrm{dpa\,s^{-1}}$ over 40 years (350 400 h), with all model parameters at their calibrated values. The bar at the end of the calculation is the depth range of the shallow intergranular oxidation damage, 2 to 3 µm, reported for the bolt by Gussev et al. [20].

Figure 9 shows the penetration depth calculated for the baffle-former bolt under concurrent irradiation and oxidation (Sec. 2.3.3). The depth increases monotonically and at a decreasing rate. After 40 years and 40 dpa the calculated depth is 2.61 µm, inside the range of 2 to 3 µm that Gussev et al. [20] reported for the shallow intergranular oxidation damage of the bolt.

4 Discussion

4.1 Oxygen Transport across and along the Intergranular Oxide

The penetration depth is set by the rate at which oxygen reaches the front, and the model resolves this rate into two steps that the single effective diffusivity of earlier grain-boundary oxidation models [26, 41, 44] does not sepa-

rate: transport along the boundary, with $D_{\mathrm{O}}^{\mathrm{eff},y}$, and delivery across the oxide to the interface, with $D_{\mathrm{O}}^{\mathrm{eff},x}$ combined with the interfacial rates through the eigenvalue μ of Eq. (22). The two diffusivities weight the same profile $D_{\mathrm{O}}(x)$ differently. The arithmetic mean is dominated by the most permeable part of the section, the harmonic mean by the least permeable. In the irradiated boundaries the porous oxide formed from the segregated plane therefore carries the transport along the boundary, whereas the delivery to the interface has to cross the dense Cr-rich oxide formed later from metal of matrix composition and is limited by it. The closure also fixes the regime of the interfacial reaction. With the calibrated values of Tables 3 and 4, the Damköhler number $kL/D_{\mathrm{O}}^{\mathrm{eff},x}$ of Sec. 2.2 passes unity when the oxide reaches a thickness of the order of 10 nm, comparable to the oxide width at half the penetration depth (Fig. 5c). The two regimes therefore coexist along the oxidized length: the interface is reaction-limited near the front, where the oxide is thin, and the consumption of oxygen in the thicker oxide behind it is increasingly limited by delivery across the oxide. The interfacial rate constants thus enter the kinetics both directly, near the front, and through the composition of the oxide they produce.

This is why the dose dependence of the penetration depth is a transport effect. In the sequential protocol the only quantity that irradiation hands to the oxidation stage is the composition of the boundary, Eq. (41); the rate constants and the oxygen supply at the mouth are the same at every dose, so the increase of depth with dose (Fig. 6) arises through $D_{\mathrm{O}}(x)$ alone, by the two terms of Eq. (16): a lower Cr_2O_3 fraction raises the volume-fraction average, and the dissolution of SiO_2 adds porosity.

4.2 Composition and Porosity across the Intergranular Oxide

Because the oxide keeps the composition it had on formation and grows from the boundary plane outward, the position across the section records the time of formation, and the segregation profile is transcribed into the oxide in the order in which the front passes through it. Two features of the measured oxide follow from this transcription without being targets of the calibration, which uses only the composition of the unirradiated oxide, the Ni content at its interface and the penetration depths (Secs. 2.4.2 and 2.4.3); the composition profiles of the irradiated specimens are predictions. The first

is the Cr valley at the centre of the oxide in the irradiated specimens, whose depth increases with dose (Figs. 4 and 5a): it is the Cr minimum of the segregation profile, oxidized first at the boundary plane and then overgrown by oxide formed from the less depleted metal on either side. The second is the location of the porosity. The Si peak of the segregation profile is transcribed to the same position, where its oxidation and the dissolution of SiO_2 convert it into the porosity concentrated within a few nanometres of the mid-plane (Fig. 5c). The porous interface that Kuang et al. [27] observed along the centre of the intergranular oxide of proton-irradiated 316L, up to the oxide tip, and attributed to the same sequence of Si oxidation and dissolution, is thus the Si counterpart of the Cr valley measured by Deng et al. [11]: in the model both are images of the segregation profile, and their position at the centre is the position of the original boundary plane.

The transcription requires that the elements not consumed at the front do not accumulate ahead of it. In the local-equilibrium limit of Eq. (35), $v_i^{\mathrm{r}} \to \infty$, the ordering of the oxygen affinities, Si ¿ Cr $\gg$ Fe $\gg$ Ni, would deplete Cr and Si and enrich Fe and Ni strongly in the metal ahead of the front, and the composition at the interface would no longer be that of the segregation profile; the scans show neither Cr depletion nor Fe enrichment on the metal side (Fig. 4). The nonequilibrium capture of Eqs. (33)–(36) [36, 43] resolves this: the Fe and Ni swept by the front are carried into the oxide and leave along its grain boundaries to the surface, where they feed the outer oxide particles that Deng et al. [10] observed to grow faster on irradiated specimens. The measured Ni content of the oxide, 1 to 4 at.% against zero in the model, indicates that part of the captured Ni is retained in the oxide rather than transported out of it, and the calculated Ni content of the metal at the interface after oxidation, 12.5 to 14.5 at.% against the measured 20 to 22 at.% reported by Deng et al. [11], that the rejected fraction of Ni is larger than the calibrated $v_{\mathrm{Ni}}^{\mathrm{r}}$ gives.

4.3 Contributions of Cr Depletion and Si Enrichment

Irradiation depletes Cr and enriches Si at the boundary together, so the experiments cannot attribute the faster intergranular oxidation to one or the other; the model can, by resetting one segregation profile at a time (Fig. 7). Under the present parameterization the Si enrichment alone produces 43 nm of the 60 nm increase of penetration depth between 0 and 3.0 dpa, reproducing the measured depth at 0.5 dpa and falling 17 nm short of it at 3.0 dpa,

whereas the Cr depletion alone adds 7 nm. The difference lies in how each element enters $D_{\mathrm{O}}(x)$. The Cr depletion lowers the Cr content of the central oxide by about 12 percentage points over the dose range (Fig. 5a) and moves the volume-fraction average of Eq. (16) part of the way from the diffusivity of one oxide toward that of the other, in a part of the section that is a few nanometres wide. The Si enrichment converts a boundary Si content that rises to 3.0 at.% (Fig. 3c) into a porosity of up to 22 vol% at the same position (Fig. 5c); even reduced by the Archie exponent, this porosity has a diffusivity orders of magnitude above that of either oxide, and the central oxide becomes the path along which oxygen reaches the front. The two routes act on the same part of the section and reinforce each other, which is why their combination produces 11 nm more than the sum of their separate contributions.

The share assigned to each route is conditional on the treatment of Si: on the complete dissolution of SiO_2, on the porosity diffusivity, which is calibrated, and on the Archie exponent, which is not. Partial retention of SiO_2, or a less connected pore network, would reduce the Si share and raise the Cr share; the ordering, however, holds as long as the dissolved SiO_2 leaves connected porosity, because the Cr route can at most raise the diffusivity of the central oxide to that of Fe_3O_4, whereas the Si route opens a path with a diffusivity set by the porosity itself. The ordering agrees with the experimental assessments that place Si above Cr. Kuang et al. [27] found that segregated Si is oxidized preferentially at the front and dissolves, leaving the porous interface discussed above, while the oxide tip was not depleted in Cr, and judged the Si effect rather than Cr depletion to be a key element of the IASCC mechanism in PWR primary water. Du et al. [13], with unirradiated model alloys in which Si and Cr were varied independently, found that the susceptibility to crack initiation rose strongly with Si content whereas a low Cr content added little, with deeper intergranular oxidation in the high-Si alloys. Those are statements about cracking susceptibility and oxide-tip chemistry; the present calculation gives the corresponding statement for the oxidation kinetics. It also qualifies the attribution of the faster intergranular oxidation of irradiated 304NG to the lower Cr content of the oxide [9, 11]: the lower Cr content at the centre of the oxide is reproduced, but in the model it accounts for a minor part of the acceleration.

4.4 Interpretation of the Measured Penetration Kinetics

The two experimental series available for 304NG differ in their apparent kinetics. The apparent effective diffusivity $\bar{D}_{\mathrm{eff}} = L^2/2t$ of the 0.5 dpa specimen exposed for 1500 h [10] exceeds that of the 3.0 dpa specimen exposed for 500 h [11] (Supplementary Information, Table S1), although the material, irradiation and exposure conditions are nominally the same and the model predicts the opposite ordering for the same boundary. The oxidation parameters were calibrated on the slower series, and a model calibrated on it cannot reproduce the absolute depths of the faster one, which it underestimates by about a factor of two (Fig. 8a). What survives the difference between the series is the relative effect of dose: the ratio between the two doses at 1500 h, 0.81 against 0.84 (Fig. 8b). The time dependence of the penetration cannot be tested against this series, because its two exposure times do not resolve when the penetration began (Sec. 3.3); the comparison with Deng et al. [10] therefore validates the effect of dose on the penetration depth, not its time exponent.

4.5 Concurrent Irradiation and Oxidation in Service

The in-service calculation is a prediction without an adjusted parameter. The parameters were calibrated on proton-irradiated 304NG exposed for 500 h, with penetration depths of 40 to 105 nm; the bolt is a neutron-irradiated type 316 steel exposed for 350 400 h, at a dose rate about 190 times lower and with irradiation and oxidation acting together. The calculation extends the calibration by a factor of about 700 in time and 25 in depth, and the calculated depth of 2.61 µm falls within the 2 to 3 µm measured for the shallow damage (Fig. 9). The measured oxide also shows the two processes on which the model rests. The original grain boundary inside the in-service oxide is depleted in Cr and in Si and acts as a path along which Fe and Ni leave toward the outer oxide [29]; in the model these are the Cr valley at the centre of the oxide, the dissolution of the oxidized Si, and the removal of the unoxidized Fe and Ni through the oxide (Sec. 4.2).

4.6 Uncertainties of the Model

The uncertainties of the model lie in the description of the oxide, in the interfacial redistribution, and in the scope of the protocol; each is paired below with the measurement or extension that would reduce it.

The oxide keeps the composition it had on formation, so a redistribution of cations within the oxide is excluded; composition profiles across the oxide at several doses would establish whether a cation diffusion term is needed. The porosity enters through an Archie exponent taken from continuum percolation of three-dimensional porous media [6, 24]; neither the exponent nor the percolation threshold of a porous intergranular oxide has been measured, the threshold depends on the shape of the pores [6], and fitted transport coefficients of reacting porous media are known to absorb kinetic effects [19]. The porosity diffusivity is calibrated for the same reason. A direct measurement of the centreline porosity at a known dose, of the kind that Kuang et al. [27] obtained by under-focus imaging, would turn the porosity from a calculated into a measured quantity and separate the exponent from the diffusivity. The complete dissolution of SiO_2 is a limiting case; the Si content of the oxide, which the model sets to zero, measures how far the real oxide departs from it. The delivery of oxygen to the interface is closed by the fundamental mode of an equivalent uniform oxide with the series resistance of the section (Supplementary Information, Sec. S3); the deviation of the lateral profile from a cosine for a nonuniform $D_{\mathrm{O}}(x)$ is neglected.

The redistribution velocities are calibrated rather than derived from the interfacial exchange diffusivities that define the diffusive velocity in the solute-capture models [4, 43], and the capture carries all unoxidized Fe and Ni out of the oxide, whereas the measured oxide retains 1 to 4 at.% Ni. A retained fraction would require the capture to be split between retention and transport through the oxide, with the Ni content of the oxide as the additional observable. On the segregation side, the Si–interstitial coupling is absorbed into an effective $D^{\mathrm{I}}_{\mathrm{Si}}$ (Sec. 2.4.1), and the calibrated profiles overestimate the Cr depletion and underestimate the Si enrichment at 0.5 dpa (Fig. 3); a dose-dependent error in the input profiles propagates directly into the oxidation stage.

The model is calibrated only in the sequential irradiation-then-exposure protocol, and the oxygen supply through the surface film is represented by the Robin condition of Eq. (42) with an assumed transfer coefficient, to which the depths are insensitive. The extension to concurrent irradiation

and oxidation requires no additional parameters (Sec. 2.3.3); what it does not include is the effect of irradiation on the oxide itself, by radiolysis of the water in the channel or by enhanced transport in the oxide, which would have to be added before predictions under in-reactor conditions are attempted.

5 Conclusions

A model that couples radiation-induced segregation to grain-boundary oxidation through the composition-dependent oxygen transport of the intergranular oxide has been developed for 304 stainless steel in PWR primary water and compared with measured segregation profiles, oxide compositions and intergranular penetration depths. The main conclusions are as follows.

1. The model closes the chain from the segregated boundary composition to the advance of the oxidation front. The segregation model, with the boundary as a point-defect sink of finite efficiency, supplies the dose-dependent Cr, Ni and Si profiles; the oxidation model consumes the segregated metal through composition-dependent interfacial reactions, records the composition of the oxide across the section as it grows, and computes from that record the two effective oxygen diffusivities that govern transport along the boundary and delivery across the oxide to the interface, with the porosity left by SiO_2 dissolution entering through Archie's law. The delivery to the interface is closed by the fundamental mode of the diffusion equation in the cross-section, which spans the reaction-limited and diffusion-limited regimes without a prescribed transport resistance.

2. The interfacial parameters are fixed by the composition of the unirradiated oxide and the Ni content of the metal at its interface, and the three transport parameters by the penetration depths after 500 h at 0, 0.5 and 3.0 dpa, with a single parameter set. Outside the calibration the model reproduces the Cr, Ni and Si concentrations at the boundary at 3.0 dpa, the composition profiles across the oxide of the irradiated specimens, and, in an independent exposure series, the ratio between doses after 1500 h. The depths of that series after 1500 h are underestimated by about a factor of two; the Ni enrichment of the metal at the interface is underestimated.

3. Because the oxide keeps its composition of formation and grows from the boundary plane outward, the segregation profile is transcribed into the oxide: the Cr minimum of the profile becomes the Cr valley at the centre of the oxide that deepens with dose, and the Si peak becomes the porosity concentrated along the centreline. Both follow from the segregation profile and the nonequilibrium capture of the unoxidized metal without being fitted, and both have been observed, the Cr valley on 304NG and the centreline porosity on proton-irradiated 316L.

4. The dose dependence of intergranular oxidation is carried entirely by the inherited segregation profile and acts through the transport resistance of the oxide rather than through the interfacial rate constants. Resetting one segregation profile at a time separates the two routes, which the experiments cannot: under the present parameterization the Si enrichment produces 43 nm of the 60 nm increase of penetration depth between 0 and 3.0 dpa and the Cr depletion 7 nm, the remainder arising from their combination in the same central part of the oxide. The ordering is conditional on the complete dissolution of SiO_2 and on the porosity transport law, and it agrees with the experimental assessments that place Si above Cr.

5. The main uncertainties are the exclusion of cation redistribution within the oxide, the porosity transport law and the complete-dissolution limit for SiO_2, and the calibrated redistribution velocities; each can be constrained by a measurement identified above.

6. Applied without adjustment to concurrent irradiation and oxidation of a type 316 baffle-former bolt at a constant dose rate over 40 years of service, the model gives a penetration depth of 2.61 µm at 40 dpa, within the 2 to 3 µm reported for the shallow intergranular oxidation damage of the bolt.

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